

Project: LifeSci360 - Clinical Research & Laboratory Management Platform

1. Introduction

LifeSci360 is a web-based Clinical Research and Laboratory Management Platform designed for pharmaceutical companies, CROs (Contract Research Organizations), and research laboratories to unify study protocols, sample tracking, patient enrollment, and compliance reporting. The platform consolidates trial data, laboratory workflows, and regulatory documentation to provide end-to-end visibility, streamlined operations, and adherence to GxP/HIPAA standards.

It supports RESTful APIs with Angular/React frontends, and backend implementations in Java Spring Boot or ASP.NET Core. The architecture emphasizes reliability, secure handling of sensitive data, and compliance with life sciences regulations.

Actors / Users

- **Research Scientist:** Manages study protocols, sample tracking, and test results.
- **Lab Technician:** Records sample processing, updates test outcomes.
- **Clinical Trial Manager:** Oversees study progress, site performance, and compliance.
- **Regulatory Officer:** Reviews documentation, compliance reports, and audit logs.
- **Data Manager:** Manages trial data, monitors quality, and prepares reports.
- **Admin:** Configures protocols, sites, users, and integrations.

2. Module Overview

2.1 Identity & Access Management

Authentication, RBAC, audit trails.

2.2 Protocol & Study Management

Study protocol registry, site setup, investigator assignments.

2.3 Sample & Laboratory Management

Sample registration, tracking, and test result recording.

2.4 Patient & Subject Management

Enrollment workflows, visit scheduling, and subject records.

2.5 Compliance & Audit Management

Audit logs, regulatory reporting, and compliance dashboards.

2.6 Analytics & Reporting

Operational dashboards, KPIs, and compliance reports.

2.7 Notifications & Alerts

Real-time alerts for study milestones, lab results, and compliance deadlines.

3. Architecture Overview

- **Frontend:** Angular or React SPA for dashboards and workflows.
- **Backend:** RESTful services (Spring Boot / ASP.NET Core).
- **Database:** Relational DB (SQL Server/MySQL/PostgreSQL).
- **Integration:** ERP, EHR/EMR systems, and regulatory portals.
- **Security:** TLS, encryption at rest, RBAC, audit logging.

4. Module Wise Design

4.1 Identity & Access Management Module

Features

- User provisioning and role assignment
- Secure login with RBAC
- Session management and password policies
- Immutable audit logs

Entities

- **User**(UserID, Name, Role, Email, Phone)
- **AuditLog**(AuditID, UserID, Action, Timestamp)

4.2 Protocol & Study Management Module

Features

- Protocol registry and lifecycle management
- Study site setup and investigator assignment
- Protocol search and retrieval

Entities

- **Protocol**(ProtocolID, Title, Phase, StartDate, EndDate, Status)
- **Site**(SiteID, Name, Location, InvestigatorID, Status)

4.3 Sample & Laboratory Management Module

Features

- Sample registration and tracking
- Laboratory test result recording

- Sample lifecycle management

Entities

- **Sample**(SampleID, ProtocolID, PatientID, CollectedDate, Status)
- **LabResult**(ResultID, SampleID, TestType, ResultValue, Date, Status)

4.4 Patient & Subject Management Module

Features

- Patient enrollment workflows
- Visit scheduling and tracking
- Subject record management

Entities

- **Patient**(PatientID, Name, DOB, ContactInfo, EnrollmentStatus)
- **Visit**(VisitID, PatientID, ProtocolID, VisitDate, Status, Notes)

4.5 Compliance & Audit Management Module

Features

- Audit log review
- Regulatory reporting generation
- Compliance dashboards

Entities

- **ComplianceReport**(ReportID, Scope, Metrics, GeneratedDate)

4.6 Analytics & Reporting Module

Features

- Study progress dashboards
- KPI tracking (Enrollment Rate, Sample Processing Rate, Compliance Score)
- Trend analysis

Entities

- **KPIReport**(ReportID, Scope, Metrics, GeneratedDate)

4.7 Notifications & Alerts Module

Features

- Real-time alerts for study milestones and lab results
- Escalation workflows
- Configurable notification channels (in-app, email, SMS)

- Notification history

Entities

- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

5. Deployment Strategy

- **Local / Dev:** Angular/React UI, Spring Boot/.NET backend, local RDBMS.
- **Staging:** Containerized services with synthetic study and sample data.
- **Production:** On-premise or hosted deployment with API gateway, managed RDBMS with replicas.
- **High Availability:** Multi-server deployments, automated backups, disaster recovery playbooks.

6. Database Design

Tables

- **User**(UserID, Name, Role, Email, Phone)
- **AuditLog**(AuditID, UserID, Action, Timestamp)
- **Protocol**(ProtocolID, Title, Phase, StartDate, EndDate, Status)
- **Site**(SiteID, Name, Location, InvestigatorID, Status)
- **Sample**(SampleID, ProtocolID, PatientID, CollectedDate, Status)
- **LabResult**(ResultID, SampleID, TestType, ResultValue, Date, Status)
- **Patient**(PatientID, Name, DOB, ContactInfo, EnrollmentStatus)
- **Visit**(VisitID, PatientID, ProtocolID, VisitDate, Status, Notes)
- **ComplianceReport**(ReportID, Scope, Metrics, GeneratedDate)
- **KPIReport**(ReportID, Scope, Metrics, GeneratedDate)
- **Notification**(NotificationID, UserID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Research Scientist Dashboard:** Protocols, samples, lab results.
- **Lab Technician Console:** Sample processing, test result entry.
- **Clinical Trial Manager Dashboard:** Study progress, site performance, compliance.
- **Regulatory Dashboard:** Audit logs, compliance reports.
- **Data Manager Console:** Trial data, quality monitoring, reporting.
- **Admin Console:** Protocol setup, site configuration, user/role management.

8. Non-Functional Requirements

- **Performance:** Handle up to **15,000 concurrent users**; sample registration latency under **300 ms**.
- **Security:** RBAC, TLS, encryption at rest, HIPAA/GxP compliance, immutable audit logs.
- **Scalability:** Horizontal scaling for study and lab services.
- **Availability:** Target **99.9% uptime** for study and lab services.
- **Maintainability:** Modular APIs, versioning, automated migrations, CI/CD pipelines.
- **Observability:** Centralized logs, metrics, and alerting.
- **Compliance:** Configurable retention policies, audit trails, regulatory reporting formats.

9. Assumptions & Constraints

- **Phase 1 scope:** Protocol registry, sample tracking, patient enrollment, lab result recording, and dashboards.
- **Excluded in Phase 1:** AI-based trial optimization, IoT device integrations, advanced mobile app features.
- **Data availability:** Historical trial and lab data available for integration.
- **Integration:** ERP, EHR/EMR systems, and regulatory portals via APIs.
- **Implementable:** Using Java Spring Boot or ASP.NET Core, Angular/React, and SQL Server/MySQL/PostgreSQL.